

Software Test Plan (STP) - Bank Management System

Project: Bank Management System

Version: 1.0

Author: Adarsha E (PES1UG24AM334)

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Status: Complete

1. Introduction

Purpose: This document defines the test plan for Bank Management System v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers customer and account management, authentication, deposits, withdrawals, balance and statements, transfers, ledger, audit, and reporting. The exclusions are listed in section 4.

References: Bank Management System SRS v1.0, Jira Backlog, Software Architecture and Design traceability.

Definitions: SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), PIN (Personal Identification Number), CLI (Command Line Interface), paise (integer unit of currency).

2. Test Items

- Account and customer module
- Authentication and authorisation module
- Deposit, withdrawal, and transfer module
- Balance, statement, ledger, and reporting module
- Validation and file persistence module
- CLI and C++17 build

3. Features to be Tested

Features mapped to SRS requirement IDs:

- BMS-F-001 to BMS-F-006: Customer records and account lifecycle
- BMS-F-010 to BMS-F-014: Authentication, logout, and role control
- BMS-F-020 to BMS-F-022: Deposit and journal entry
- BMS-F-030 to BMS-F-033: Withdrawal and account limits
- BMS-F-040 to BMS-F-042: Balance and statements
- BMS-F-050 to BMS-F-052: Atomic fund transfer
- BMS-F-060 to BMS-F-064: Ledger, reconciliation, audit, and reporting

- BMS-NF-001 to BMS-NF-007: Performance, reliability, and build quality
- BMS-SR-001 to BMS-SR-007: Credential, data, input, and access security

4. Features Not to be Tested

- ATM hardware, card readers, cash dispensers, and kiosk interfaces
- Inter-bank settlement, cheque clearing, loans, and card issuance
- Internet and mobile banking interfaces and statutory reporting

5. Test Approach / Strategy

Levels:

- Unit tests (validation, money arithmetic, and service methods)
- Integration tests (transactions with persistence and ledger)
- System tests (end-to-end CLI workflows)
- Acceptance tests (Customer, Teller, and Manager use cases)

Types:

- Functional and boundary-value testing
- Regression and recovery testing before and after COMMIT and between transfer legs
- Performance testing (latency and 10,000-account startup)
- Usability and security testing

Entry Criteria: Relevant module built, test data and environment ready, expected results defined; journal format and failure hooks agreed.

Exit Criteria: All planned cases executed, high-priority requirements pass, no critical or major defect open, and reconciliation passes.

5.1 Security Validation

- Verify salted credential hashes and no plaintext or terminal echo
- Check role restrictions in the service layer
- Test invalid input, long strings, and monetary overflow
- Verify owner-only files, append-only logs, and rejection of a second instance
- Inject write and disk-full failures; check COMMIT replay, account CRC recovery, and reconciliation

6. Test Environment

Hardware: Standard desktop or laptop; record specifications for performance tests.

Software: Bank Management System C++17 console application on Linux and Windows.

Tools: g++, MinGW, unit-test harness, scripted CLI tests, Jira for defects.

Test Data: Synthetic Savings and Current accounts, ACTIVE/LOCKED/CLOSED states, transaction histories, and boundary amounts.

7. Test Schedule

Milestones:

- Test plan and case design: 02-Oct-2026

- Environment setup: Sprint 1, before execution
- Unit and integration execution: Sprints 1 to 3 as modules merge
- System and regression execution: Sprint 4
- UAT and summary report: After full regression in Sprint 4

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual and automated)
- Test Scripts
- Synthetic Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Adarsha E	Prepare plan, coordinate and record testing
Account and Auth Lead	Dhanush S	Support account, authentication, and validation tests
Transaction Lead	Vidit Soni	Support transaction and persistence tests and fixes
Review Team	Team 11	Review results and release readiness

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early builds and run module tests as code merges
Incomplete or repeated recovery	Fail before/after COMMIT and between transfer legs; verify once-only replay
Corrupt record or full disk	Corrupt an account CRC and simulate disk full; verify recovery or no state change

11. Assumptions & Dependencies

- The SRS v1.0 and its requirement IDs remain the test baseline
- Journal format, COMMIT rules, and test hooks are approved before integration
- Synthetic accounts and a stable build are available before execution

12. Suspension & Resumption Criteria

Suspend testing if:

- A build or environment failure blocks execution
- Recovery or reconciliation fails on clean test data

Resume testing if:

- Blocking defects are fixed and the build is stable
- Failed cases and affected regression cases pass

13. Test Case Management & Traceability

The SRS section 8 RTM maps each requirement to a test case. TC-REL-01 checks incomplete batches, once-only replay, and account-record recovery. Examples:

- BMS-F-011 (account lockout) -> TC-AUT-02
- BMS-F-050 (fund transfer) -> TC-TRF-01
- BMS-NF-002 (crash-safe journal) -> TC-REL-01

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Sprint execution status
- Final Test Summary Report